Joined VSL stack: analytical review addendum

2026-10-21 — JM1; no new traffic law or numerical experiment

Meaning of the result. The complete remaining-time objective is more than a formal infinite rollout with a conditional chain rule. For this fixed model it admits a common finite representation over the allowed advice interval and has rational formulas (ratios of polynomials in advice) between finitely many possible switching values. This does not make it globally smooth or establish a useful information or optimizer advantage.

1. Same model and assumptions

Use the unchanged joined contract, sections 1–7: fixed $N = 10$ rows, continuous stocks $0 \le n_{i\ell} \le J = 15$, common FD, selected H budgets, normalized requests, positive equal nominal priorities, capped sharing, one forward fallback, and one-row forward or diagonal advancement. Final E can only merge into A's outlet. All movements use old stock. The initial cohort is independent of advice; there are no later arrivals or blocked downstream outlets. Advice is one constant $v \in I = [30, 60]$ km/h, with the stated time/space bounds. Empty donors skip undefined ratios. The fixed member's preferences and budgets use finitely many rational formulas, comparisons and min/max branches.

Let $h_i = N - i + 1$, $M(n) = \sum_{i,\ell} n_{i\ell}$ and

$$V(n) = \sum_{i,\ell} h_i n_{i\ell}, \qquad \eta = \Delta x \rho_c = 15/4, \qquad c(v) = v\Delta t/\Delta x \in [1/2, 1]. \tag{1}$$

Here V is mass-weighted remaining row count, not a speed. Every realized movement, including an exit, reduces V by its mass.

2. The progress argument supplies a uniform finite prefix

Regard desired fractions b as independent elements of their fixed compact allowed simplices, keeping terminal E-to-A compulsory. For an empty donor, any such b represents the same zero requests. Define $\Gamma(n, b, v)$ as the sum of all realized boundary movement amounts in one update. The budget/allocation laws give

$$V(n^+) = V(n) - \Gamma(n, b, v), \qquad M(n) \le V(n) \le NM(n). \tag{2}$$

The expanded map Γ is continuous in (n, b, v): budgets and requests are continuous, capped sharing has unique continuous flow amounts at demand caps and zero supply, and fallback takes minima of continuous residuals. This statement does not assume continuity of the native incentive map $b(n)$.

If $V > 0$, take the furthest occupied row. Its downstream row is empty or is the free outlet. Each allowed destination there has positive receiving budget, each positive donor has positive sending, and its requests sum to that sending. Positive priorities therefore serve some positive mass. Hence $\Gamma > 0$ whenever $V > 0$, for every admissible b.

On the compact set of physical stocks, allowed b, $v \in I$ and $V \ge \eta$, continuity and positivity give a minimum $\mu > 0$. As V has a finite uniform upper bound, (2) implies a finite common bound on steps to $V < \eta$, independently of the native branch sequence or advice. Then $M < \eta$; no inflow can undo this. The existing contract's free-tail proof applies: all requests are served, each donor advances $c(v)n_{i\ell}$, and the remaining value is exact. Choose one integer K_* beyond this common entry bound. For every $v \in I$,

$$\mathcal{J}(v) = \Delta t \sum_{k=0}^{K_*-1} M(n^k(v)) + \frac{\Delta x}{v} V(n^{K_*}(v)). \tag{3}$$

No numerical estimate of μ or simulated trajectory is used here. This is an existence argument, not an efficient clearance algorithm or a claim of exact finite emptying.

3. Finite branch formulas imply regular advice intervals

Fix the admitted initial state. On a chosen branch, density lookahead is linear, incentive ratios are rational wherever evaluated, geometry terms are constant, H budgets are min/max choices of rational terms, and donor normalization is rational. For capped sharing, fix fully served requests H and entitlement-limited requests U. In the constrained branch,

$$\lambda = \frac{r - \sum_{a \in H} m_a}{|U|}, \qquad z_a = m_a \ (a \in H), \quad z_a = \lambda \ (a \in U), \quad |U| > 0. \tag{4}$$

The unconstrained/zero cases have their explicit finite formulas. Fallback and stock updates preserve rationality. There are finitely many choices per update and K_* is finite, so each candidate trace gives rational states, branch conditions and full value (3) as functions of v.

Every nonidentically-zero rational comparison has finitely many roots and denominator zeros on I: its numerator and denominator are polynomials of finite degree. Identically zero comparisons are persistent ties, not an infinite collection of separate switches; undefined-ratio branches are excluded by their own conditions. Taking the finite union over candidate traces yields a finite exceptional set. On each remaining open interval, the selected formulas are fixed and $\mathcal{J}$ is rational with nonzero denominators, hence smooth. Values at exceptional points may jump or have unequal one-sided slopes. No global minimum-attainment claim follows.

For such an interval, write the selected step formula as $F^{[k]}$ and $s^k = \partial n^k / \partial v$. The contract's derivative is justified:

$$\begin{aligned} s^0 &= 0, \qquad s^{k+1} = F_n^{[k]} s^k + F_v^{[k]}, \\ \mathcal{J}_v &= \Delta t \sum_{k < K_*} \mathbf{1}^\top s^k + \frac{\Delta x}{v} h^\top s^{K_*} - \frac{\Delta x}{v^2} V(n^{K_*}). \end{aligned} \tag{5}$$

The three value terms concern prefix traffic, residual-state sensitivity and direct residual travel time. A persistent state tie need not admit a global state Jacobian: the selected smooth formula along the advice curve is what is differentiated. Ordinary derivatives can still be zero. A finite partition need not be cheaply enumerable. No AD/FD verification, smoothing, continuous-time saltation or continuum-limit result is asserted.

4. A bounded negative information result in the free regime

If $M(n^0) < \eta$, the entire episode already lies in the exact-tail regime for every allowed v. Therefore

$$\mathcal{J}(v, n^0) = \frac{\Delta x}{v} V(n^0), \qquad \mathcal{J}_v = -\frac{\Delta x}{v^2} V(n^0). \tag{6}$$

For any nonempty cohort, maximum advice uniquely minimizes this expression. If two initializations share row totals but differ in lane allocation, they share V and the whole objective for every advice. Hidden lane allocation then has zero action value. The empty cohort has zero value for all advice.

The existing X/Z contrast has $M_0 = 38.25 > \eta$ and therefore is not covered by this sufficient free-regime condition. Its different local services do not prove different full-objective optimum advice. Outside the free regime, fixed physical receiving/ceiling budgets and state-dependent preferences prevent treating advice as a universal time rescaling, but no general positive information-gap or maximum-advice dominance theorem is proved here.

Review boundary. These are deductions from the selected finite, closed-cohort member. Continuing inflow, blocked exits, changed action timing, heterogeneous-FD mapping or different traffic laws need their own qualification. No source calibration, numerical gradient correctness, optimizer advantage, physical transfer, full-lane necessity or discrete-order necessity follows. The earlier model and its equations remain unchanged.